

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10 - PRESSURE

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have finished the unit on pressure. Now you will write a Final Explanation to answer our Driving Question: "Why did the jerrycan crush itself, and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" Answer each section in the space under its prompt. Use evidence from your lessons, your Driving Question Board, your cumulative model and your calculations. Where a prompt asks for a calculation, show the formula, the substitution and the unit. Where a prompt asks for a diagram, label every arrow. Write in full sentences and use the words: pressure, force, area, atmospheric pressure, density, depth, Pascal's Principle.

Section 1: Why Did the Jerrycan Crush Itself?

In Lesson 1 you watched a sealed, empty 20-litre jerrycan crumple inward after a small amount of air was removed from it, while nobody touched it. Write a complete explanation of why it collapsed. (a) Describe what the air inside and outside the jerrycan was doing BEFORE air was removed and say why the jerrycan stayed intact. (b) Explain what changed when air was removed. (c) A jerrycan is strong enough to carry 20 kg of water every day without deforming, yet this sealed, empty jerrycan collapsed. Explain how both facts can be true. (d) Draw and label a pressure-balance diagram showing the jerrycan before and after air removal. (e) Use evidence from the collapsing tin (Lesson 4), the Magdeburg Hemispheres and the mercury

(a) Before air was removed, the air inside pushed outward on the walls and the atmosphere outside pushed inward on the walls. The pressures were equal ($P_{\text{outside}} = P_{\text{inside}}$), so the net force on each wall was zero and the jerrycan kept its shape. (b) When air was removed, the pressure inside fell. Atmospheric pressure outside, about 101.3 kPa (101,325 Pa) at sea level, stayed the same, so $P_{\text{outside}} > P_{\text{inside}}$. The unbalanced net inward force on every wall is $F = \Delta P \times A$, because pressure is force per unit area ($P = F/A$). This pushed the flexible plastic walls inward on all sides. When the cap was opened, air rushed back in, the pressures balanced again and the jerrycan partly recovered. (c) Both facts are true because the jerrycan is not weak. Carrying 20 kg of water is a different kind of load: the water's weight is carried mainly by the base and handle, and the water pushes outward on the walls, so the walls are not squeezed inward. The plastic is strong against that load. In the sealed, empty jerrycan, removing air lowered the inside pressure, so the atmosphere pushed inward on every wall with nothing balancing it. The atmosphere is very powerful because it is a fluid with weight, and it pushes in ALL directions on every surface. The jerrycan collapsed not because it was weak, but because the usual balance of pressures was lost. (d) Diagram: Before: equal arrows pointing inward (outside air) and outward (inside air) on every wall, labelled $P_{\text{outside}} = P_{\text{inside}}$, net force = 0. After: long arrows pointing inward from outside, short arrows pointing outward from inside, labelled $P_{\text{outside}} > P_{\text{inside}}$, net force inward. (e) Evidence: the metal tin was sealed after steam had been generated inside and then cooled in cold water, and it imploded dramatically. This was a replay of the jerrycan event: the tin collapsed because the pressure inside became lower than the atmospheric pressure outside, so the net force from the atmosphere was inward. The Magdeburg Hemispheres could not be pulled apart by teams of horses because the atmosphere pushed them together once the air inside was removed. The mercury barometer holds up about 760 mm of mercury using atmospheric pressure alone. Together these show that atmospheric pressure is real, large and acts in all directions.

barometer.	
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Section 2: Why Don't Our Bodies Get Crushed?

Atmospheric pressure acts on your body all the time, yet you are not crushed. (a) Using what you learned about pressure balance, explain why not. (b) Explain what is the same and what is different between your body and the jerrycan after air was removed. (c) Explain why atmospheric pressure is described as acting in ALL directions, and give one piece of evidence from our lessons.	(a) Our bodies are filled with fluids and tissues that push outward with a pressure equal to the atmospheric pressure pushing inward. The pressures are balanced, so the net force on our body surfaces is zero and we are not crushed. (b) Same: the same atmospheric pressure, about 101.3 kPa, acts on both. Different: for the body the inside pushes back equally, whereas in the jerrycan the inside pressure was lowered, so $P_{\text{outside}} > P_{\text{inside}}$ and the net force pushed the walls inward. (c) The atmosphere is a fluid, and pressure in a fluid pushes on a surface from every side, not only from above. Evidence: the jerrycan crumpled inward on ALL sides, and the Magdeburg Hemispheres were held together by air pushing on them from all directions.
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Section 3: Pressure in Liquids and Kenyan Structures

In Lessons 3, 6 and 8 you investigated liquid pressure. (a) State the three factors that control liquid pressure and describe the evidence from the water-jet bottle investigations that showed how depth and density matter. (b) Write the formula and calculate the pressure at the base of the Sasumua Dam reservoir (depth about 40 m, fresh water density 1000 kg/m³, $g = 10 \text{ N/kg}$). (c) Calculate the pressure due to the seawater on a diver at 10 m depth in the Indian Ocean off Mombasa (seawater density 1025 kg/m³). (d) Compare your answers in (b) and (c) with atmospheric pressure. What does the comparison tell you about the design of the Sasumua Dam wall?	(a) Liquid pressure depends on depth (h), the density of the liquid (ρ) and gravitational field strength (g). In the bottle investigation, the deepest hole made the water jet travel the farthest, so pressure increases with depth. The salt-water jet travelled farther than the fresh-water jet from the same depth, so a denser liquid gives more pressure. (b) $P = \rho gh = 1000 \times 10 \times 40 = 400,000 \text{ Pa}$ (400 kPa). (c) $P = \rho gh = 1025 \times 10 \times 10 = 102,500 \text{ Pa}$. (d) Atmospheric pressure is about 100,000 Pa (101,325 Pa). The 102,500 Pa from the seawater alone at 10 m is about the same size as the whole atmosphere that crushed the jerrycan (in fresh water, 10 m of depth gives about 100,000 Pa). A diver also has the atmosphere pushing on the water surface, so the total pressure on the diver is about 203,825 Pa. At the base of Sasumua Dam the pressure due to the water alone (400,000 Pa) is about four times atmospheric pressure. Because pressure increases with depth, the force on the dam wall is greatest at the bottom, so engineers must use calculated pressure values to choose a wall strong enough, especially near the base, and not rely on rough guesses.
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Section 4: Pascal's Principle and Hydraulic Machines

In Lessons 5 and 7 you used connected syringes and watched a hydraulic car lift in a Nairobi garage. (a) State Pascal's Principle in your own words. (b) Explain how a small force on a small piston can lift a heavy vehicle such as a Toyota Noah. (c) A small piston has an area of 5 cm^2 and you push it with a force of 40 N . The large piston has an area of 500 cm^2 . Calculate the force on the large piston. (d) Name two other Kenyan hydraulic machines we studied. (e) Explain why removing air bubbles from the syringe system was important.

(a) Pressure applied to an enclosed (confined) fluid is transmitted equally and undiminished in all directions. (b) Pushing the small piston creates a pressure $P = F_1/A_1$. This pressure is transmitted through the fluid to the large piston, where the same pressure acts on a much larger area, so $F_2 = P \times A_2$ is much larger than F_1 . The machine multiplies force by using a large output area and a small input area. (c) $F_1/A_1 = F_2/A_2$, so $F_2 = F_1 \times A_2/A_1 = 40 \times 500/5 = 4,000 \text{ N}$. (d) Matatu brake systems and the sugarcane presses in Mumias (also hospital beds at Kenyatta National Hospital). (e) Air bubbles can be squashed, so some of the push would be used compressing air instead of being transmitted through the fluid. Removing them lets the pressure transmit reliably.

Section 5: Putting It All Together

Write a final answer to the Driving Question: "Why did the jerrycan crush itself, and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" (a) Say how the jerrycan collapse is DIFFERENT from a hydraulic machine, and say what they have in common. (b) Choose TWO Kenyan examples (for instance a dam, a diver, a matatu brake, a syringe or the Kenya Meteorological Department barometer) and explain which pressure idea each uses. (c) Name one question from our Driving Question Board that you can now answer, and one that is still open.

(a) Different: the jerrycan collapsed because of an uncontrolled pressure difference ($P_{\text{outside}} > P_{\text{inside}}$) that produced a net inward force, $F = \Delta P \times A$. A hydraulic machine uses a controlled transmission of pressure through an enclosed fluid. Common: both depend on pressure being force per unit area ($P = F/A$), and both involve fluids pushing on surfaces. (b) Example 1: Sasumua Dam uses liquid pressure, $P = \rho gh$. Pressure is greatest at the base, so the wall must be strongest there. Example 2: matatu hydraulic brakes use Pascal's Principle. The driver's small force on the brake pedal creates pressure that is transmitted equally through the brake fluid to larger pistons, which multiplies the force and stops the vehicle. (Other valid examples: the Mombasa diver feels $P = \rho gh$ rising with depth; the KMD barometer measures atmospheric pressure of about 101.3 kPa .) (c) Answered: why did the jerrycan crush? Because removing air made the atmospheric pressure outside greater than the pressure inside. Open (future inquiry): a valid example is how atmospheric pressure changes with altitude, or how pressure behaves in moving fluids. Unanswered questions are not failures but the starting point of future inquiry.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the anchoring phenomenon (jerrycan)	Clearly explains the pressure imbalance before and after air removal, uses P_{outside} and P_{inside} , links to net force ($F = \Delta P \times A$), and uses a correct, fully labelled diagram with evidence from the tin, the Magdeburg Hemispheres and the barometer.	Explains that outside pressure became greater than inside pressure and the jerrycan was crushed, with a diagram and some evidence, but with minor gaps in reasoning or labelling.	States that the air crushed the jerrycan without explaining the pressure difference or the balanced state before, or gives little or no evidence.
Use of formulas and calculations	Correctly selects $P = F/A$, $P = \rho gh$ and $F_1/A_1 = F_2/A_2$, shows substitution and units, and obtains 400,000 Pa, 102,500 Pa and 4,000 N.	Uses the correct formulas with mostly correct answers, with minor errors in units or in showing the working.	Chooses incorrect formulas or makes major errors, or gives answers without working or units.
Use of evidence from lessons	Draws specifically on the water-jet investigations, the collapsing tin, the Magdeburg Hemispheres, the barometer, the syringes and the car lift to support each claim.	Refers to some lesson evidence to support most claims.	Makes claims with little or no reference to lesson evidence.
Connecting concepts and Kenyan applications	Distinguishes uncontrolled pressure difference from controlled pressure transmission, links atmospheric pressure, liquid pressure and Pascal's Principle, and explains two Kenyan examples accurately.	Links most concepts and explains at least one Kenyan example accurately.	Treats ideas separately, with applications that are missing or inaccurate.
Scientific communication	Writes clear, logical, complete sentences with correct terms (pressure, force, area, density, depth, Pascal's Principle) and reflects thoughtfully on the Driving Question Board.	Writes generally clear explanations with mostly correct scientific terms.	Writing is unclear or incomplete, with frequent misuse of terms.

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